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Summary — Data Science graduate with hands-on experience building end-to-end ML pipelines, predictive models, ETL workflows, and interactive dashboards. Skilled in Python, Flask, React, SQL and Scikit-learn, with real-world projects deployed in production, including a capstone project developed in partnership with Nouvelair Tunisie.

Skills

- **Languages & Tools:** Python, R, Bash, Java, C#, Git, VS Code, PyCharm
- **Data & Databases:** Pandas, NumPy, SQL, PostgreSQL, MySQL, MongoDB, Linux, ETL, Data Pipelines
- **Machine Learning:** Regression, Classification, Clustering, XGBoost, LSTM (TensorFlow/Keras), Hyperparameter Tuning, Model Evaluation, SHAP
- **Data Engineering:** REST APIs, Flask, FastAPI, Docker, AWS (EC2, S3), Data Cleaning, Feature Engineering
- **Visualization:** Matplotlib, Seaborn, Dashboards, React.js, Recharts, Power BI, Tableau
- **Web Development:** Flask, React.js, Django, Laravel, JavaScript, HTML/CSS

Technical Achievements

- Developed multiple end-to-end ML pipelines (preprocessing → model → API → dashboard).
- Built and deployed production-ready REST APIs for ML models using Flask.
- Reduced data processing time by **35%** through optimized feature engineering.
- Designed scalable data pipelines handling **10k+ records/day**.
- Deployed full-stack applications on cloud (AWS EC2) with Docker containers.

ML & Data Science Projects

FuelTrack – Intelligent Aircraft Fuel Tracking & Optimization Platform (Capstone Project)

Partners: Nouvelair Tunisie & ATDS **Tools:** Flask, PostgreSQL, React, XGBoost, LSTM, SHAP, RAG (Groq LLaMA 3.3-70B), Power BI, Docker

- Designed and built an intelligent web platform to track, analyze, and predict aircraft fuel consumption and CO₂ emissions, replacing a manual workflow based on LIDO Flight, PDC FlightOps and Excel OFP files.
- Built predictive models (XGBoost, Gradient Boosting, Random Forest, LSTM) for Block Fuel, Trip Fuel, cost and CO₂ estimation, with registration-based bias correction (HFC).
- Implemented a RAG module (Groq LLaMA 3.3-70B, Gemini 2.5 Flash fallback) for conversational assistance based on SQL-aggregated flight data.
- Built a full-stack Flask/PostgreSQL/React architecture with JWT authentication and role-based access (admin, analyst, pilot, viewer), following a hybrid SCRUM + CRISP-DM methodology over 6 sprints.

AI Email Assistant – NLP Short Reply Generator (Python, Flask)

Tools: Python, Flask, TextBlob, NLTK, Bootstrap, REST API, Git

- Developed an intelligent email reply generator with sentiment analysis and intent detection.
- Built a full NLP engine (30+ templates) with REST API and responsive UI.

Football Data Pipeline & Analytics Dashboard (Python, Pandas)

Tools: Python, Pandas, NumPy, Scikit-learn, Matplotlib, Flask, Docker, REST API, Git

- Processed 20k+ match records into a complete ETL pipeline with predictive features.
- Built interactive dashboards and deployed statistics API using Flask + Docker.

Human Resource Management System (Python, MySQL)

Tools: Python, MySQL, SQLAlchemy, Pandas, Git

- Built a full HR system with employee management, attendance, and analytics modules.
- Automated SQL workflows and generated performance reports for 500+ employees.

AI-Powered School Attendance System (Flask, MongoDB, React)

Tools: Flask, MongoDB, React, Node.js, Axios, Scikit-learn, Docker, Git

- Designed ML-driven backend processing 5,000+ attendance events per day.
- Created real-time dashboards in React, improving reporting efficiency by 40%.

Global Health Data Engineering Pipeline (Python, Streamlit)

Tools: Python, Pandas, NumPy, Streamlit, Plotly, Scikit-learn, Git

- Developed a full data engineering pipeline to clean, unify and analyze global health indicators (life expectancy, infant mortality, HALE).
- Built an interactive Streamlit dashboard with choropleth maps, trends, comparisons, and dynamic filters.

Experience

Nouvelair Tunisie — IT Technology & Projects Department Capstone Intern — Data Scientist / Full-Stack Developer - Designed and built FuelTrack, an intelligent web platform to track, analyze, predict and optimize aircraft fuel consumption, developed in partnership with Nouvelair Tunisie and ATDS. - Implemented Machine Learning and Deep Learning models (XGBoost, LSTM) to predict fuel consumption and CO₂ emissions, with bias correction by aircraft registration. - Built a full-stack architecture (Flask, PostgreSQL, React) integrating a RAG module for decision support. - Followed a hybrid SCRUM + CRISP-DM methodology under the supervision of Ms. Leila Ghorbel (academic supervisor) and Mr. Ramzi Fazzi and Mr. Ghayth Ajra (industry supervisors).	Feb 2026 – Jun 2026
Elite Council Consulting Full-Stack Developer Intern — Data/AI Module - Built backend pipelines for AI attendance system handling thousands of daily events. - Developed and deployed React dashboards for real-time analytics. - Implemented Flask/MongoDB modules improving data flow reliability by 25%.	May 2024 – Aug 2024
CEFI — Centre de Formation Informatique Python Programming Instructor - Taught Python to 30+ students, focusing on data structures and mini-analytics projects. - Designed practice labs, assignments, and data manipulation exercises.	Feb 2024 – Jun 2024

Education

Professional Master’s in Data Science — FSM, Faculty of Sciences of Monastir <i>Graduated July 6, 2026</i> <i>Relevant Courses:</i> Machine Learning, Data Mining, Big Data Analytics, AI Systems <i>Capstone Project:</i> FuelTrack – Intelligent Aircraft Fuel Tracking & Optimization Platform, in partnership with Nouvelair Tunisie and ATDS	2024–2026
Bachelor’s Degree in Computer Science — Software Engineering — FSM, Faculty of Sciences of Monastir <i>Graduated in May 2024</i> <i>Relevant Coursework:</i> Software Engineering, Database Systems, Object-Oriented Programming <i>Capstone Project:</i> Human Resource Management System (HRMS) — Developed a full-stack web application to streamline HR operations, including employee management, attendance tracking, leave management, recruitment, and performance evaluation through a centralized and user-friendly platform.	2021–2024

Certifications

- Python for Data Science, AI & Development — IBM
- AWS Cloud Practitioner (Foundations) — AWS
- Full-Stack Web Development (React, Flask, MongoDB) — ECC
- Python Programming Certification — JNL Center

Soft Skills

- Analytical thinking & problem-solving
- Rapid learner, adaptable to new tools
- Teamwork & clear communication
- Time management, reliability, attention to detail

Languages

- Arabic:** Native
- English:** Professional proficiency
- French:** Intermediate